Frozen self-supervised features and temporal context for free-living activity recognition on CAPTURE-24

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Abstract

Wrist accelerometers are widely used to measure daily activity, but recognising activities in free-living data remains hard. We study the 10-class task on CAPTURE-24, labelling 10 s windows for 151 participants. We first reimplement the published random forest + HMM baseline (held-out Cohen's κ 0.66 against 0.64 published) and describe two evaluation pitfalls: splitting HMM sequences at annotation gaps leaks label information, and balanced random forests with too few trees leave many windows without out-of-bag predictions. We also find that annotators leave a gap at almost every activity change, so all scored windows lie inside bouts. A frozen self-supervised encoder followed by a bidirectional GRU over about ten minutes of windows reaches held-out κ 0.79, 0.10 above our random forest + HMM and 0.04 above the same encoder with HMM smoothing. Macro-F1 does not improve over the latter, and part of the κ gain comes from how model outputs are converted to classes.

1 Introduction

Wrist accelerometers are worn in large cohort studies to measure how people spend their day. Turning the signal into activities requires labelled free-living data, and CAPTURE-24 (Chan et al., 2024) provides it: participants wore a wrist device for about a day while a wearable camera photographed their activity every 20 to 40 seconds. The dataset paper reports a random forest and a convolutional network, each followed by a hidden Markov model (HMM), on a fixed split that holds out participants P101 to P151.

The task labels each 10 s window, but the activities themselves, such as cooking, a commute or an afternoon at a desk, last minutes. The 10-class scheme also separates activities that look alike from the wrist, such as household chores and manual work. Two ways to close that gap are a model that reads minutes of signal at once and an encoder pretrained on far more data than CAPTURE-24 (Yuan et al., 2024). We test both, after first checking the classical baseline and its evaluation protocol.

The best model, a frozen encoder with a bidirectional GRU, reached held-out κ 0.79 (Table 2). Our contributions are:

1. A 10-class evaluation of the `ssl-wearables` encoder, frozen and fine-tuned, with an HMM and with a recurrent context model, and a decomposition of the gain into encoder, context and decision rule.
2. A reimplementation of the published random forest + HMM and two evaluation pitfalls, including an approximate closed-form rule for missing out-of-bag output in balanced random forests.
3. Label gaps in CAPTURE-24: only 46 of 609,580 adjacent labelled windows differ in activity, so all scored windows lie inside bouts.

2 Related work

Willetts et al. (2018) introduced the balanced random forest + HMM for wrist accelerometry in UK Biobank and the 10-class scheme we use. Chan et al. (2024) released CAPTURE-24 with random forest and CNN baselines and listed self-supervised learning as future work. Yuan et al. (2024) pretrained a ResNet encoder on 700,000 person-days of UK Biobank data (Doherty et al., 2017) and reported CAPTURE-24 results for



Figure 1: Six hours of held-out participant P101: camera labels, the random forest + HMM (RF-HMM) and the frozen encoder with a bidirectional GRU (SSL-BiGRU), on labelled windows only.

4 intensity classes. Their tool ActiNet (Acquah et al., 2025) classifies the same 4 classes on 30 s windows with this encoder, an HMM and a sleep correction, evaluated by cross-validation over all participants. The first version of the ActiNet paper froze the encoder and trained a final layer, while the current version fine-tunes all layers; its authors name recurrent and transformer models as future work. Haresamudram et al. (2022) compared self-supervised methods for activity recognition across datasets, and Ordóñez and Roggen (2016) combined convolutional and LSTM layers to model temporal context. As of this writing, we found no published 10-class result for the `ssl-wearables` encoder on CAPTURE-24.

3 Data and protocol

Each recording (Chan Chang et al., 2021) is 100 Hz triaxial acceleration in g with an annotation per sample. We cut recordings into non-overlapping 10 s windows and dropped windows with fewer than 1,000 samples or any missing value. A window took the most frequent class among its labelled samples, and stayed unlabelled when "unlabelled" was its most frequent annotation. Our labelled window counts then equal the paper's, 922,199 in total. Unlabelled windows were kept as model input but never trained on or scored.

Model development used 5-fold cross-validation on P001 to P100, grouped by participant and stratified on the labelled windows. Each final model was fitted on P001 to P100 and scored on P101 to P151. We report macro-F1, the unweighted mean of per-class F1, and Cohen's κ (Cohen, 1960). The code uses scikit-learn (Pedregosa et al., 2011), imbalanced-learn (Lemaître et al., 2017) and PyTorch (Paszke et al., 2019).

Held-out intervals are 95% percentile intervals from a participant-level bootstrap with 2,000 draws (Field and Welsh, 2007), computed from per-participant confusion matrices, and differences between models are paired on the same draws. Over the project the held-out set was scored six times: an earlier version of RF-HMM that split sequences at annotation gaps, RF-HMM, the reimplemented paper pipeline and the three encoder models. We report intervals without correcting for this repeated use of the held-out set.

3.1 Label gaps

Annotators leave an unlabelled gap at almost every change of activity. Among 609,580 adjacent pairs of labelled development windows, only 46 are two different activities, and nearly all label changes are into or out of "unlabelled". An HMM that counts transitions between adjacent labelled windows therefore learns a matrix close to "never switch", with most off-diagonal entries at zero. No score here measures behaviour at a change of activity.

4 Reproduction and pitfalls

Our random forest (RF) was imbalanced-learn's balanced random forest (Chen et al., 2004) with 300 trees on 39 hand-made features per window: moments, quantiles, autocorrelation, spectral and peak features of the acceleration magnitude, plus per-axis means, standard deviations and correlations. RF-HMM smoothed the forest's probability vector with an HMM (Rabiner, 1989) whose prior and transitions came from the training participants. Because each tree trains on a class-balanced sample, we treated the forest's output

Table 1: Reproduction. Held-out scores on P101 to P151 with 95% participant-bootstrap intervals, and cross-validation means on P001 to P100. The last row is a paired held-out difference.

Model	Cross-validation Macro-F1	Cross-validation κ	Held out Macro-F1	Held out κ
Paper RF-HMM, published			0.486	0.637
Paper RF-HMM, reimplemented	0.517	0.670	0.535 [0.490, 0.566]	0.660 [0.630, 0.692]
RF	0.406	0.535	0.415 [0.380, 0.438]	0.530 [0.503, 0.555]
RF-HMM	0.522	0.708	0.566 [0.524, 0.595]	0.690 [0.653, 0.725]
RF-HMM − Paper RF-HMM			+0.031 [−0.011, 0.080]	+0.029 [0.001, 0.058]

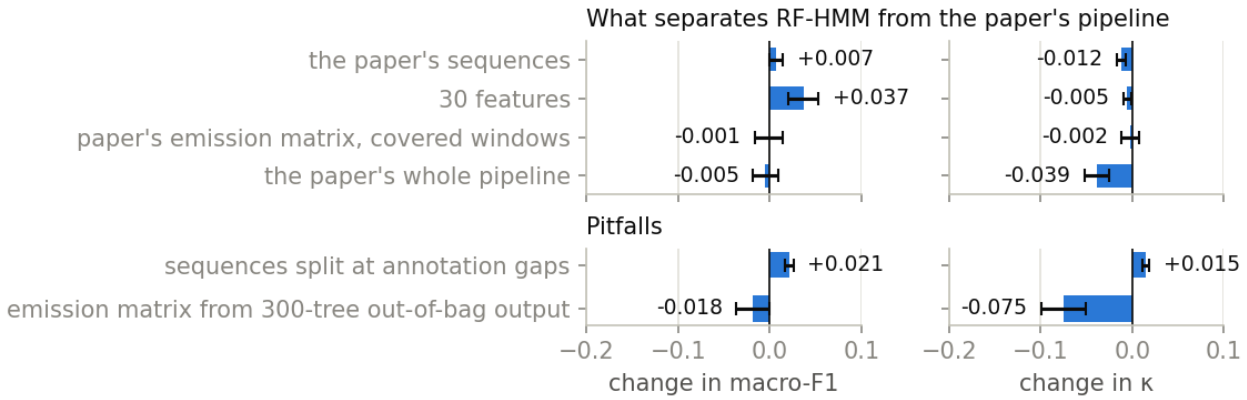


Figure 2: RF-HMM with one choice swapped: change against RF-HMM on the five cross-validation folds (mean and standard error). Upper block: the paper's choices. Lower block: the two pitfalls.

as proportional to $P(\text{window} \mid \text{activity})$ and used it as the emission likelihood. RF-HMM decoded every window of a recording with Viterbi and was scored on labelled windows.

We reimplemented the paper's pipeline (Paper RF-HMM) from its code, with its configuration fixed before any score was seen. It used 3,000 trees on 30 features, an emission matrix averaged from out-of-bag forest output, a uniform prior, and decoding of labelled windows only. In Table 1 the published κ lies inside the reimplementation's interval and the published macro-F1 sits just below it. RF-HMM is slightly above the reimplementation, and we use it as the classical baseline.

To see which of the paper's choices matter, we swapped them into RF-HMM one at a time on the cross-validation folds. No single choice explains the gap: each changes κ by at most 0.012 (Figure 2), while the whole pipeline is 0.039 below RF-HMM.

The first pitfall is a leak. If sequences are split wherever annotation stops, the gaps tell the HMM where each activity starts and ends, which raised macro-F1 by about 0.02 on our folds (Figure 2). A deployed model would not have that information. The paper's protocol decodes labelled windows only, a mild form of the same leak that did not help on these folds.

The second pitfall concerns out-of-bag output in balanced forests. In imbalanced-learn, a window is out-of-bag for a tree if it is in that tree's undersampled pool and the tree's bootstrap missed it. For a class with n_c training windows, where the smallest class has $n_{\min}$, a window enters the pool with probability about $1 - e^{-n_{\min}/n_c}$ and is then missed with probability about $1/e$. With T trees, the share of the class without any out-of-bag output is approximately

$$u_c = \left(1 - \frac{1 - e^{-n_{\min}/n_c}}{e}\right)^T. \tag{1}$$

With 300 trees, 29% of sleep and sitting training windows got no out-of-bag output, as the rule predicts, and with 3,000 trees none did. The library fills those windows with rows of zeros and only warns, and an emission matrix fitted on that output cost about 0.08 κ (Figure 2). Leaving uncovered windows out, or fitting the matrix out-of-fold, removes the loss.

Table 2: Main results. Cross-validation means (networks averaged over three seeds) and held-out scores with 95% participant-bootstrap intervals. Lower rows are paired held-out differences, each model at the rule fixed before scoring. The published CNN + HMM was trained from scratch and is not paired.

	Cross-validation		Held out	
Model	Macro-F1	κ	Macro-F1	κ
CNN + HMM, published			0.576	0.735
RF-HMM	0.522	0.708	0.566 [0.524, 0.595]	0.690 [0.653, 0.725]
SSL-FT-HMM	0.578	0.741	0.619 [0.557, 0.659]	0.742 [0.702, 0.778]
SSL-HMM	0.603	0.776	0.638 [0.571, 0.681]	0.750 [0.717, 0.783]
SSL-BiGRU	0.580	0.814	0.642 [0.577, 0.677]	0.790 [0.758, 0.819]
SSL-BiGRU − RF-HMM			+0.076 [0.021, 0.111]	+0.100 [0.076, 0.124]
SSL-BiGRU − SSL-FT-HMM			+0.022 [−0.025, 0.057]	+0.048 [0.026, 0.071]
SSL-BiGRU − SSL-HMM			+0.004 [−0.025, 0.026]	+0.040 [0.023, 0.057]

5 Models

The encoder is `harnet10` from `ssl-wearables` (Yuan et al., 2024; OxWearables, 2024), an 18-layer 1D ResNet pretrained without labels on UK Biobank data; CAPTURE-24 was not used in pretraining. It reads 10 s at 30 Hz, so we resampled each window to 300 samples, and it outputs 1,024 features per window.

SSL-FT-HMM fine-tuned all encoder layers with a linear head, using the encoder authors' settings where given (Adam with AMSGrad, learning rate 10^{-4}, batches of 1,000 windows), with one recipe and early stopping on a fifth of the training participants. Its softmax was divided by the training class prior to give scaled likelihoods (Bourlard and Morgan, 1994) and decoded with the RF-HMM's HMM. It followed the current version of ActiNet.

SSL-BiGRU passed every window once through the frozen encoder and fed the features of consecutive windows to a bidirectional GRU (Cho et al., 2014; Schuster and Paliwal, 1997) with 2 layers and 128 units per direction, and dropout 0.5. It trained on chunks of 64 windows (about ten minutes), with unlabelled windows as input but no loss on them, and at prediction it read each whole recording. The control, SSL-HMM, applies the same network to each window alone and smooths its output with the HMM. It followed the first version of ActiNet, with this network as the head in place of a single layer. To separate encoder from context, we trained the same two networks on the 39 hand-made features (Feat-HMM and Feat-BiGRU) with unchanged settings.

Network settings, the candidate models and the selection rule were written down before any of their scores were seen. Networks were trained with three seeds per fold. For SSL-BiGRU, we compared its top class with its prior-divided output through the HMM and chose the higher cross-validation macro-F1, with κ deciding ties within 0.005. The two tied, and the top class won on κ. Each selected model was fitted once on P001 to P100 with seed 42. SSL-HMM was scored on the held-out participants after the SSL-BiGRU result was known, with its configuration fixed in advance.

The decision rule turns a model's output into a class. We scored each model with its output divided by the class prior raised to α (Saerens et al., 2002), for $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. At $\alpha = 0$ the class frequencies are kept, which favours common activities; at $\alpha = 1$ they are removed, which favours rare ones. The rules used for the held-out scores were fixed before scoring: SSL-BiGRU used $\alpha = 0$ and the HMM models used $\alpha = 1$.

6 Results and decomposition

SSL-BiGRU reached κ 0.79 on the held-out participants (Table 2), 0.10 above RF-HMM. Against SSL-HMM it gained 0.04 κ and had the higher κ for 42 of 51 participants, while their macro-F1 was level. The published CNN + HMM is below SSL-BiGRU's κ interval, but it was trained from scratch and is not paired with our models.

Most errors are far from label changes. Windows within 30 s of a label change are only 8% of the windows, and leaving them out raised RF-HMM's cross-validation κ from 0.71 to 0.74. Two thirds of its errors lie more

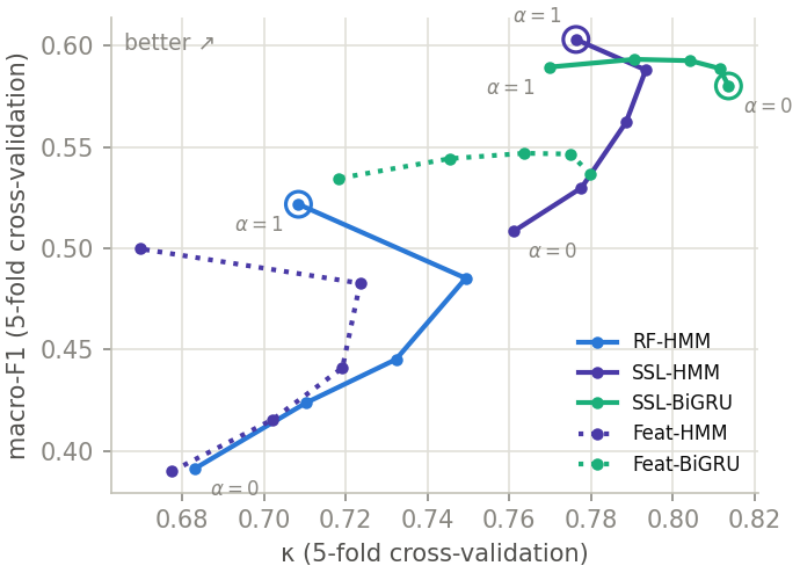


Figure 3: Cross-validation macro-F1 against κ as the decision rule moves from class frequencies kept ($\alpha = 0$) to removed ($\alpha = 1$). Rings mark the rule used for the held-out scores; dotted lines are the networks on hand-made features.

than a minute from any change, which suggests stretches of a few minutes that a 10 s classifier misreads as a whole. Context helps there: on windows 1 to 5 minutes from a change, the error rate fell from 0.25 for SSL-HMM to 0.21 for SSL-BiGRU. Adding an HMM after SSL-BiGRU lowered its κ, since the network had already removed the flicker.

The frozen encoder was at least as good as the fine-tuned one, both in cross-validation and on the held-out participants (Table 2). The two models also differ in their heads, and only one fine-tuning recipe was tried.

Part of every comparison is the decision rule. SSL-BiGRU keeps the class frequencies in its output while SSL-HMM divides them out before the HMM, so the two favour different metrics (Figure 3). At each model's best rule, SSL-BiGRU's cross-validation κ gain over SSL-HMM falls from 0.04 to 0.02, and its cross-validation gain over RF-HMM falls from 0.11 to 0.06. Comparing the four networks at their best rules suggests that the encoder and context overlap: the encoder adds 0.07 κ without context and 0.03 with it, and context adds 0.06 on hand-made features and 0.02 on the encoder. The best macro-F1 at any rule was 0.60.

7 Discussion and limitations

A frozen encoder pretrained on another cohort, with a small recurrent model over minutes of windows, gave the highest held-out κ we measured on the 10-class task. The gain is in κ: SSL-BiGRU's macro-F1 differences from SSL-FT-HMM and SSL-HMM both include zero, and manual work and mixed activity stayed near or below F1 0.3 in cross-validation.

The evaluation has known weak points. The held-out gaps compare models at different decision rules, selection used the same cross-validation folds we report, only one seed was scored held out, and the held-out set was used six times over the project. The data come from one UK cohort wearing one device, rare activities rest on few participants (only 8 of the 51 held-out participants did any sport), and the encoder's authors also built the dataset, so its design may favour this benchmark. Finally, SSL-BiGRU, like Viterbi decoding, needs the whole recording, so it works offline only.

Data labelled continuously through changes of activity would show how much of each score depends on the label gaps. In practice, a frozen pretrained encoder with sequence context is a strong default for this task, and comparisons should report the decision rule alongside the score. The code, notebooks and every reported number are available at `https://github.com/Alexr951/Capture-24`.

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